

A Literal Counterexample to “Non-Weak-Embedding Implies Expander”

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Status

This packet gives a likely valid counterexample to the literal wording of the last open question in Arzhantseva–Tessera [1]: a bounded-degree sequence of finite connected graphs need not be an expander merely because it does not weakly embed into ℓ^2 .

The result should be read as a wording/normalization counterexample. It does not address the more interesting strengthened question in which one rules out linear-size expander pieces or asks for a structural characterization modulo such pieces.

Source Problem Evidence

The source question appears in Section 8, “Open problems”, on page 20 of the arXiv PDF:

cannot coarsely embed.

- It results from (2.1), that expanders (or even large subsets of expanders) do not weakly embed into ℓ^p for all $1 \leq p < \infty$. On the other hand all of our examples do weakly embed into ℓ^2 (just take a coarse embedding of Q_n). Therefore, it seems relevant to ask whether a sequence of finite graphs with uniformly bounded degree that does not weakly embed into ℓ^2 is necessarily an expander.

Transcription

After observing that expanders, and even large subsets of expanders, do not weakly embed into ℓ^p , while the paper’s relative expander examples do weakly embed into ℓ^2 , the authors ask whether a sequence of finite graphs with uniformly bounded degree that does not weakly embed into ℓ^2 is necessarily an expander.

Definitions

A sequence of maps $f_n : X_n \rightarrow \ell^2$ is a weak embedding if there is a constant K such that every f_n is K -Lipschitz and, for every fixed $R > 0$,

$$\lim_{n \rightarrow \infty} \sup_{x \in X_n} \frac{|f_n^{-1}(B(f_n(x), R))|}{|X_n|} = 0.$$

An expander sequence is a sequence of finite connected graphs of uniformly bounded degree, with $|X_n| \rightarrow \infty$, whose Cheeger constants are bounded below by a positive constant.

Proof Intuition

The obstruction to weak embedding is inherited from any linear-size expander subgraph. If a graph X_n contains an isometric copy of an expander E_n with $|E_n| \geq c|X_n|$, then any uniformly Lipschitz map $X_n \rightarrow \ell^2$ collapses a positive proportion of E_n , hence a positive proportion of X_n , into one bounded Hilbert ball. But this has nothing to do with the global Cheeger constant of X_n . Attaching a long path to E_n by one edge destroys expansion while preserving the linear-size expander obstruction to weak embedding.

Lemma 1. *Let (E_n) be a bounded-degree expander sequence. For every $K > 0$ there are $R > 0$ and $c > 0$ such that, for every sequence of K -Lipschitz maps $f_n : E_n \rightarrow \ell^2$, for all n sufficiently large there is a vertex $v_n \in E_n$ with*

$$|f_n^{-1}(B(f_n(v_n), R))| \geq c|E_n|.$$

Proof. The usual expander Poincare inequality gives a constant B , depending only on the expander family and its degree bound, such that every K -Lipschitz map $f : E_n \rightarrow \ell^2$ satisfies

$$\frac{1}{|E_n|^2} \sum_{x, y \in E_n} \|f(x) - f(y)\|^2 \leq BK^2.$$

This is precisely the $p = 2$ form of the expander inequality recalled in Section 2 of [1]. Therefore some vertex $v \in E_n$ satisfies

$$\frac{1}{|E_n|} \sum_{y \in E_n} \|f(v) - f(y)\|^2 \leq BK^2.$$

By Markov's inequality, at least half of the vertices $y \in E_n$ obey $\|f(v) - f(y)\| \leq \sqrt{2B}K$. Take $R = \sqrt{2B}K$ and $c = 1/2$. $\square$

Theorem 1. *There exists a sequence (X_n) of finite connected graphs with uniformly bounded degree such that:*

1. (X_n) is not an expander sequence;
2. (X_n) does not weakly embed into ℓ^2 .

Proof. Let (E_n) be any bounded-degree expander sequence, and write $m_n = |E_n|$. Let P_n be a path with m_n vertices. Choose a vertex $a_n \in E_n$ and an endpoint $b_n \in P_n$, and form X_n by joining a_n and b_n with one new edge.

The degrees of X_n are uniformly bounded. Also X_n is connected and $|X_n| = 2m_n \rightarrow \infty$.

The sequence (X_n) is not an expander. Indeed, let A_n be the terminal half of the path P_n , away from the attachment point. Then $|A_n| \leq |X_n|/2$, while only one edge leaves A_n . Hence

$$h(X_n) \leq \frac{1}{|A_n|} \rightarrow 0.$$

It remains to prove that (X_n) does not weakly embed into ℓ^2 . The copy of E_n inside X_n is isometric: the attached path creates no shortcut between two vertices of E_n . Suppose $f_n : X_n \rightarrow \ell^2$

is a sequence of K -Lipschitz maps. Restrict f_n to E_n . By Lemma 1, there are $R > 0$, $c > 0$, and vertices $v_n \in E_n$ such that, for infinitely many n and in fact eventually,

$$|f_n^{-1}(B(f_n(v_n), R)) \cap E_n| \geq c|E_n|.$$

Consequently

$$\sup_{x \in X_n} \frac{|f_n^{-1}(B(f_n(x), R))|}{|X_n|} \geq \frac{c|E_n|}{2|E_n|} = \frac{c}{2}$$

along the same tail. Thus the weak-embedding condition fails for every uniform Lipschitz constant K . Therefore (X_n) does not weakly embed into ℓ^2 . $\square$

Verification Notes

No computation is needed. The proof reduces the non-weak-embedding assertion to the standard expander Poincare inequality and uses an explicit one-edge bottleneck to force $h(X_n) \rightarrow 0$. The construction also shows a slightly more general principle: any bounded-degree graph sequence containing a linear-size isometric expander piece need not weakly embed into ℓ^2 , regardless of its global Cheeger constant.

Limitations

This packet answers only the literal formulation displayed above. It is not a new relative-expander construction, and it does not address the intended frontier suggested by [1]: understanding non-weak-embeddability once linear-size expander pieces have been excluded. A natural repaired version would ask whether a bounded-degree sequence with no linear-size expander pieces, or no large subsets satisfying the expander Poincare collapse, must weakly embed into ℓ^2 .

Novelty and Literature Check

Before promotion, I searched the run indexes for arXiv:1402.1481, “relative expanders”, “weakly embed into ℓ^2 ”, and “necessarily an expander”. No existing packet or attempt covered this literal final question. Web searches for the exact wording found the source paper and later papers answering other open problems from Section 8, including the finitely generated group problem [2] and the coarse Baum–Connes problem, but no exact prior answer to this literal graph-sequence question.

The novelty confidence is deliberately modest. The construction is elementary and the source paper itself points out that large subsets of expanders obstruct weak embedding. The value of the packet is to record a precise counterexample to the literal wording and to clarify the hypothesis needed for a more interesting version.

Human Review Recommendation

Check that the weak-embedding normalization in [1] uses $|X_n|$ in the denominator, and that $|E_n|/|X_n| = 1/2$. Then verify that the attached path does not create shortcuts inside E_n and that the terminal half of the path forces $h(X_n) \rightarrow 0$.

References

- [1] G. Arzhantseva and R. Tessera, *Relative expanders*, arXiv:1402.1481, 2014. <https://arxiv.org/abs/1402.1481>
- [2] G. Arzhantseva and R. Tessera, *Admitting a coarse embedding is not preserved under group extensions*, arXiv:1605.01192, 2016. <https://arxiv.org/abs/1605.01192>